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TEKNOLOGI
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MODUL PRAKTIKUM

Teknik Informatika

IF-9P2

Pemrograman Piranti Bergerak

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MODUL

User Interface

1. Objective

Pada latihan kali ini, diharapkan mahasiswa dapat memahami penggunaan View dan View Group pada android untuk merancang User Interface.

2. Activities

Pada latihan kali ini Anda akan mengembangkan sebuah halaman detail yang di dalamnya terdiri dari beberapa komponen View dan ViewGroup Layout.

3. Exercise

Menjalankan Aplikasi menampilkan tampilan (tampilan akan sesuai dengan apa yang dibuat di layout editor).

3.1 Basic Fragment

1. Buat Project baru di Android Studio dengan kriteria sebagai berikut:

Nama Project	MyViewProject
Target & Minimum Target SDK	Phone and Tablet, Api level 21
Tipe Activity	Empty Activity
Activity Name	MainActivity
Use AndroidX artifacts	TRUE
Language	Kotlin

2. Lalu buka berkas **build.gradle (Module: app)** dan tambahkan satu baris ini di bagian *dependencies*:

1. **implementation 'de.hdodenhof:circleimageview:3.0.0'**

Sehingga berkas **build.gradle(Module: app)** kita sekarang menjadi seperti ini:

```
1.  apply plugin: 'com.android.application'
2.  apply plugin: 'kotlin-android-extensions'
3.  apply plugin: 'kotlin-android'
4.
5.  android {
6.      compileSdkVersion 29
7.      buildToolsVersion '29.0.1'
8.      defaultConfig {
9.          applicationId "com.dicoding.picodiploma.myviewandviews"
10.         minSdkVersion 21
11.         targetSdkVersion 29
12.         versionCode 1
13.         versionName "1.0"
14.         testInstrumentationRunner
15.         "androidx.test.runner.AndroidJUnitRunner"
16.     }
17.     buildTypes {
18.         release {
19.             minifyEnabled false
20.             proguardFiles getDefaultProguardFile('proguard-
21.             android.txt'), 'proguard-rules.pro'
22.         }
23.     }
24.     dependencies {
25.         implementation fileTree(dir: 'libs', include: ['*.jar'])
26.         androidTestImplementation('androidx.test.espresso:espresso-
27.         core:3.1.0', {
28.             exclude group: 'com.android.support', module: 'support-
29.             annotations'
30.         })
31.         implementation 'androidx.appcompat:appcompat:1.1.0'
32.         implementation 'androidx.vectordrawable:vectordrawable-
33.         animated:1.1.0'
34.         testImplementation 'junit:junit:4.12'
35.         implementation "androidx.core:core-ktx:1.0.2"
36.         implementation "org.jetbrains.kotlin:kotlin-stdlib-
37.         jdk7:$kotlin_version"
```

```

34.     implementation 'de.hdodenhof:circleimageview:3.0.0'
35. }

```

3. Lalu buka berkas **strings.xml** di **res** → **values**. Sesuaikan isinya dengan seperti ini:

```

1. <resources>
2.     <string name="app_name">MyViewAndViews</string>
3.     <string name="content_text">Google officially announced its much-
        anticipated Pixel phones; the Pixel and Pixel XL, on October 4. We
        attended Google's London UK event, mirroring the main one taking place
        in San Francisco, US, where the firm unwrapped the new Android 7.1
        Nougat devices which will apparently usurp Google's long-standing Nexus
        series.</string>
4.     <string name="content_specs_display">5.0 inches\n
5.         FHD AMOLED at 441ppi\n
6.         2.5D Corning® Gorilla® Glass 4</string>
7.     <string name="content_specs_size">5.6 x 2.7 x 0.2 ~ 0.3 inches
        143.8 x 69.5 x 7.3 ~ 8.5 mm</string>
8.     <string name="content_specs_battery">2,770 mAh battery\n
9.         Standby time (LTE): up to 19 days\n
10.        Talk time (3g/WCDMA): up to 26 hours\n
11.        Internet use time (Wi-Fi): up to 13 hours\n
12.        Internet use time (LTE): up to 13 hours\n
13.        Video playback: up to 13 hours\n
14.        Audio playback (via headset): up to 110 hours\n
15.        Fast charging: up to 7 hours of use from only 15 minutes of
        charging</string>
16. </resources>

```

4. Selanjutnya, unduh asset-nya di classroom. Bila telah selesai, salin semua asset yang dibutuhkan ke dalam direktori **res** → **drawable**.

5. Buka berkas **activity_main.xml** dan ubah layout utama menjadi seperti ini:

```

1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white">
8.
9. </ScrollView>

```

6. Kemudian tambahkan layout utama di dalam **ScrollView**.

```
1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white">
8.     <LinearLayout
9.         android:layout_width="match_parent"
10.        android:layout_height="wrap_content"
11.        android:orientation="vertical">
12.
13.     </LinearLayout>
14. </ScrollView>
```

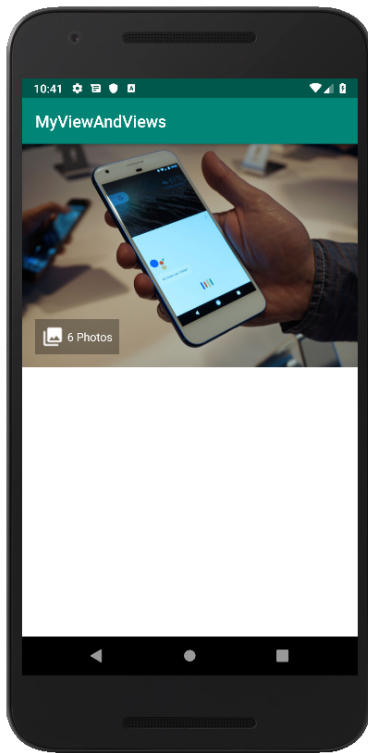
7. Kemudian kita masukkan komponen di dalam **LinearLayout** yang sudah ditambahkan sebelumnya.

```
1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white">
8.     <LinearLayout
9.         android:layout_width="match_parent"
10.        android:layout_height="wrap_content"
11.        android:orientation="vertical">
12.         <FrameLayout
13.             android:layout_width="match_parent"
14.             android:layout_height="wrap_content">
15.             <ImageView
16.                 android:layout_width="match_parent"
17.                 android:layout_height="wrap_content"
18.                 android:adjustViewBounds="true"
19.                 android:src="@drawable/pixel_google"
20.                 android:scaleType="fitXY"/>
21.             <TextView
22.                 android:layout_width="wrap_content"
23.                 android:layout_height="wrap_content"
24.                 android:padding="8dp"
25.                 android:text="6 Photos"
26.                 android:gravity="center_vertical"
27.                 android:drawableLeft="@drawable/
ic_collections_white_18dp"
28.                 android:drawablePadding="4dp"
```

```

29.         android:textAppearance="@style/
TextAppearance.AppCompat.Small"
30.         android:background="#4D000000"
31.         android:textColor="@android:color/white"
32.         android:layout_marginLeft="16dp"
33.         android:layout_marginBottom="16dp"
34.         android:layout_gravity="bottom" />
35.     </FrameLayout>
36. </LinearLayout>
37. </ScrollView>

```



8. Selanjutnya, tambahkan textview di bawah **FrameLayout**, menjadi seperti ini:

```

1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white"
8.
9.     tools:context="com.dicoding.picodiploma.myviewandviews.MainActivity">
10.     <LinearLayout
11.         android:layout_width="match_parent"
12.         android:layout_height="wrap_content"

```

```

12.         android:orientation="vertical">
13.         <FrameLayout ...>
14.
15.         <TextView
16.             android:layout_width="match_parent"
17.             android:layout_height="wrap_content"
18.             android:textSize="32sp"
19.             android:text="$735"
20.             android:layout_marginLeft="16dp"
21.             android:layout_marginRight="16dp"
22.             android:layout_marginTop="16dp"
23.             android:layout_marginBottom="8dp"
24.             android:textColor="@android:color/black"/>
25.         <TextView
26.             android:layout_width="match_parent"
27.             android:layout_height="wrap_content"
28.             android:text="Stock hanya 5 buah"
29.             android:textSize="12sp"
30.             android:layout_marginLeft="16dp"
31.             android:layout_marginRight="16dp"
32.             android:layout_marginBottom="16dp"/>
33.         <TextView
34.             android:layout_width="match_parent"
35.             android:layout_height="wrap_content"
36.             android:text="@string/content_text"
37.             android:layout_marginLeft="16dp"
38.             android:layout_marginRight="16dp"
39.             android:layout_marginBottom="16dp"
40.             android:lineSpacingMultiplier="1"
41.             android:textColor="@android:color/black"/>
42.     </LinearLayout>
43. </ScrollView>

```

9. Setelah menambahkan beberapa textview, selanjutnya kita akan menambahkan tablelayout.

```

1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white"
8.
9.     tools:context="com.dicoding.picodiploma.myviewandviews.MainActivity">
10.     <LinearLayout
11.         android:layout_width="match_parent"
12.         android:layout_height="wrap_content"
13.         android:orientation="vertical">
14.         <FrameLayout ...>
15.         <TextView ...>
16.         <TextView ...>
17.         <TextView ...>

```

```

18.         <TextView
19.             android:layout_width="match_parent"
20.             android:layout_height="wrap_content"
21.             android:text="Spesifikasi"
22.             android:textSize="12sp"
23.             android:layout_marginLeft="16dp"
24.             android:layout_marginRight="16dp"
25.             android:layout_marginBottom="8dp"/>
26.         <TableLayout
27.             android:layout_width="match_parent"
28.             android:layout_height="wrap_content"
29.             android:layout_marginLeft="16dp"
30.             android:layout_marginRight="16dp"
31.             android:layout_marginBottom="16dp">
32.             <TableRow
33.                 android:layout_width="match_parent"
34.                 android:layout_height="wrap_content"
35.                 android:layout_marginBottom="8dp">
36.                 <TextView
37.                     android:layout_width="wrap_content"
38.                     android:layout_height="wrap_content"
39.                     android:textSize="14sp"
40.                     android:layout_marginRight="16dp"
41.                     android:text="Display"/>
42.                 <TextView
43.                     android:layout_width="match_parent"
44.                     android:layout_height="wrap_content"
45.                     android:textSize="14sp"
46.                     android:layout_weight="1"
47.                     android:text="@string/content_specs_display"
48.                     android:textColor="@android:color/black"/>
49.                 </TableRow>
50.                 <TableRow
51.                     android:layout_width="match_parent"
52.                     android:layout_height="wrap_content"
53.                     android:layout_marginBottom="8dp">
54.                     <TextView
55.                         android:layout_width="wrap_content"
56.                         android:layout_height="wrap_content"
57.                         android:textSize="14sp"
58.                         android:layout_marginRight="16dp"
59.                         android:text="Size"/>
60.                     <TextView
61.                         android:layout_width="match_parent"
62.                         android:layout_height="wrap_content"
63.                         android:textSize="14sp"
64.                         android:layout_weight="1"
65.                         android:text="@string/content_specs_size"
66.                         android:textColor="@android:color/black"/>
67.                     </TableRow>
68.                 <TableRow
69.                     android:layout_width="match_parent"
70.                     android:layout_height="wrap_content"
71.                     android:layout_marginBottom="8dp">
72.                     <TextView

```



```

73.             android:layout_width="wrap_content"
74.             android:layout_height="wrap_content"
75.             android:textSize="14sp"
76.             android:layout_marginRight="16dp"
77.             android:text="Battery"/>
78.         <TextView
79.             android:layout_width="match_parent"
80.             android:layout_height="wrap_content"
81.             android:textSize="14sp"
82.             android:layout_weight="1"
83.             android:text="@string/content_specs_battery"
84.             android:textColor="@android:color/black"/>
85.     </TableRow>
86. </TableLayout>
87. </LinearLayout>
88. </ScrollView>

```

10. Kemudian setelah menambahkan tablelayout, kita masukkan viewgroup lagi. Tambahkan RelativeLayout dan button di bawah **TableLayout** menjadi seperti ini:

```

1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white"
8.     tools:context="com.dicoding.picodiploma.myviewandviews.MainActivity">
9.     <LinearLayout
10.         android:layout_width="match_parent"
11.         android:layout_height="wrap_content"
12.         android:orientation="vertical">
13.         <FrameLayout ...>
14.         <TextView ...>
15.         <TextView ...>
16.         <TextView ...>
17.         <TextView ...>
18.         <TableLayout ...>
19.
20.         <TextView
21.             android:layout_width="match_parent"
22.             android:layout_height="wrap_content"
23.             android:text="Dijual oleh"
24.             android:textSize="12sp"
25.             android:layout_marginLeft="16dp"
26.             android:layout_marginRight="16dp"
27.             android:layout_marginBottom="8dp"/>
28.         <RelativeLayout
29.             android:layout_width="match_parent"
30.             android:layout_height="wrap_content"
31.             android:layout_marginLeft="16dp"
32.             android:layout_marginRight="16dp"
33.             android:layout_marginBottom="16dp">

```

```

34.         <de.hdodenhof.circleimageview.CircleImageView
35.             android:layout_width="56dp"
36.             android:layout_height="56dp"
37.             android:src="@drawable/photo_2"
38.             android:layout_centerVertical="true"
39.             android:id="@+id/profile_image"
40.             android:layout_marginRight="16dp"/>
41.         <TextView
42.             android:layout_width="match_parent"
43.             android:layout_height="wrap_content"
44.             android:layout_toRightOf="@id/profile_image"
45.             android:text="Narenda Wicaksono"
46.             android:textColor="@android:color/black"
47.             android:layout_centerVertical="true"/>
48.     </RelativeLayout>
49.     <Button
50.         android:layout_width="match_parent"
51.         android:layout_height="wrap_content"
52.         android:text="Beli"
53.         android:layout_marginLeft="16dp"
54.         android:layout_marginRight="16dp"
55.         android:layout_marginBottom="16dp"/>
56. </LinearLayout>
57. </ScrollView>

```

11. Maka setelah setiap teks yang dipindahkan ke ***string.xml***, ***activity_main.xml*** menjadi seperti ini:

```

1. <?xml version="1.0" encoding="utf-8"?>
2. <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3.     xmlns:tools="http://schemas.android.com/tools"
4.     android:id="@+id/activity_main"
5.     android:layout_width="match_parent"
6.     android:layout_height="match_parent"
7.     android:background="@android:color/white"
8.     tools:context="com.dicoding.picodiploma.myviewandviews.MainActivity">
9.
10.    <LinearLayout
11.        android:layout_width="match_parent"
12.        android:layout_height="wrap_content"
13.        android:orientation="vertical">
14.
15.        <FrameLayout
16.            android:layout_width="match_parent"
17.            android:layout_height="wrap_content">
18.
19.            <ImageView
20.                android:layout_width="match_parent"
21.                android:layout_height="wrap_content"
22.                android:adjustViewBounds="true"

```

```

23.             android:scaleType="fitXY"
24.             android:src="@drawable/pixel_google" />
25.
26.         <TextView
27.             android:layout_width="wrap_content"
28.             android:layout_height="wrap_content"
29.             android:layout_gravity="bottom"
30.             android:layout_marginStart="16dp"
31.             android:layout_marginBottom="16dp"
32.             android:background="#4D000000"
33.             android:drawableStart="@drawable/
ic_collections_white_18dp"
34.             android:drawablePadding="4dp"
35.             android:gravity="center_vertical"
36.             android:padding="16dp"
37.             android:text="@string/dummy_photos"
38.             android:textAppearance="@style/
TextAppearance.AppCompat.Small"
39.             android:textColor="@android:color/white" />
40.     </FrameLayout>
41.
42.     <TextView
43.         android:layout_width="match_parent"
44.         android:layout_height="wrap_content"
45.         android:layout_marginLeft="16dp"
46.         android:layout_marginTop="16dp"
47.         android:layout_marginRight="16dp"
48.         android:layout_marginBottom="16dp"
49.         android:text="@string/dummy_value"
50.         android:textColor="@android:color/black"
51.         android:textSize="32sp" />
52.
53.     <TextView
54.         android:layout_width="match_parent"
55.         android:layout_height="wrap_content"
56.         android:layout_marginLeft="16dp"
57.         android:layout_marginRight="16dp"
58.         android:layout_marginBottom="16dp"
59.         android:text="@string/stock"
60.         android:textSize="12sp" />
61.
62.     <TextView
63.         android:layout_width="match_parent"
64.         android:layout_height="wrap_content"
65.         android:layout_marginLeft="16dp"
66.         android:layout_marginRight="16dp"
67.         android:layout_marginBottom="16dp"
68.         android:lineSpacingMultiplier="1"
69.         android:text="@string/content_text"
70.         android:textColor="@android:color/black" />

```

```

71.
72.     <TextView
73.         android:layout_width="match_parent"
74.         android:layout_height="wrap_content"
75.         android:layout_marginLeft="16dp"
76.         android:layout_marginRight="16dp"
77.         android:layout_marginBottom="16dp"
78.         android:text="@string/specification"
79.         android:textSize="12sp" />
80.
81.     <TableLayout
82.         android:layout_width="match_parent"
83.         android:layout_height="wrap_content"
84.         android:layout_marginLeft="16dp"
85.         android:layout_marginRight="16dp"
86.         android:layout_marginBottom="16dp">
87.
88.         <TableRow
89.             android:layout_width="match_parent"
90.             android:layout_height="wrap_content"
91.             android:layout_marginBottom="16dp">
92.
93.                 <TextView
94.                     android:layout_width="wrap_content"
95.                     android:layout_height="wrap_content"
96.                     android:layout_marginEnd="16dp"
97.                     android:text="@string/display"
98.                     android:textSize="14sp" />
99.
100.                <TextView
101.                    android:layout_width="match_parent"
102.                    android:layout_height="wrap_content"
103.                    android:layout_weight="1"
104.                    android:text="@string/content_specs_display"
105.                    android:textColor="@android:color/black"
106.                    android:textSize="14sp" />
107.            </TableRow>
108.
109.            <TableRow
110.                android:layout_width="match_parent"
111.                android:layout_height="wrap_content"
112.                android:layout_marginBottom="16dp">
113.
114.                <TextView
115.                    android:layout_width="wrap_content"
116.                    android:layout_height="wrap_content"
117.                    android:layout_marginEnd="16dp"
118.                    android:text="@string/size"
119.                    android:textSize="14sp" />
120.

```

```

121.         <TextView
122.             android:layout_width="match_parent"
123.             android:layout_height="wrap_content"
124.             android:layout_weight="1"
125.             android:text="@string/content_specs_size"
126.             android:textColor="@android:color/black"
127.             android:textSize="14sp" />
128.     </TableRow>
129.
130.     <TableRow
131.         android:layout_width="match_parent"
132.         android:layout_height="wrap_content"
133.         android:layout_marginBottom="16dp">
134.
135.         <TextView
136.             android:layout_width="wrap_content"
137.             android:layout_height="wrap_content"
138.             android:layout_marginEnd="16dp"
139.             android:text="@string/battery"
140.             android:textSize="14sp" />
141.
142.         <TextView
143.             android:layout_width="match_parent"
144.             android:layout_height="wrap_content"
145.             android:layout_weight="1"
146.             android:text="@string/content_specs_battery"
147.             android:textColor="@android:color/black"
148.             android:textSize="14sp" />
149.     </TableRow>
150. </TableLayout>
151.
152.     <TextView
153.         android:layout_width="match_parent"
154.         android:layout_height="wrap_content"
155.         android:layout_marginLeft="16dp"
156.         android:layout_marginRight="16dp"
157.         android:layout_marginBottom="16dp"
158.         android:text="@string/seller"
159.         android:textSize="12sp" />
160.
161.     <RelativeLayout
162.         android:layout_width="match_parent"
163.         android:layout_height="wrap_content"
164.         android:layout_marginLeft="16dp"
165.         android:layout_marginRight="16dp"
166.         android:layout_marginBottom="16dp">
167.
168.         <de.hdodenhof.circleimageview.CircleImageView
169.             android:id="@+id/profile_image"
170.             android:layout_width="56dp"

```

```

171.             android:layout_height="56dp"
172.             android:layout_centerVertical="true"
173.             android:layout_marginEnd="16dp"
174.             android:src="@drawable/photo_2" />
175.
176.         <TextView
177.             android:layout_width="match_parent"
178.             android:layout_height="wrap_content"
179.             android:layout_centerVertical="true"
180.             android:layout_toEndOf="@+id/profile_image"
181.             android:text="@string/my_name"
182.             android:textColor="@android:color/black" />
183.     </RelativeLayout>
184.
185.     <Button
186.         android:layout_width="match_parent"
187.         android:layout_height="wrap_content"
188.         android:layout_marginLeft="16dp"
189.         android:layout_marginRight="16dp"
190.         android:layout_marginBottom="16dp"
191.         android:text="@string/buy" />
192. </LinearLayout>
193.</ScrollView>

```

12. Terakhir, pada **MainActivity** tambahkan beberapa baris kode berikut:

```

1.  if (supportActionBar != null) {
2.      (supportActionBar as ActionBar).title = "Google Pixel"
3.  }

```

Sehingga kode yang ada di **MainActivity** menjadi seperti berikut:

```

1.  class MainActivity : AppCompatActivity() {
2.
3.      override fun onCreate(savedInstanceState: Bundle?) {
4.          super.onCreate(savedInstanceState)
5.          setContentView(R.layout.activity_main)
6.
7.          if (supportActionBar != null) {
8.              (supportActionBar as ActionBar).title = "Google Pixel"
9.          }
10.     }
11. }

```